🔷 Question 1: Accept User Input Until "exit" is Typed

📌 Question:

Write a program that keeps accepting user input until the user types `"exit"`. After exiting, print all the entered inputs.

✅ Solution (Using while loop)

```
data = []
user_input = ""

while user_input != "exit":
    user_input = input("Enter something (type 'exit' to stop): ")

    if user_input != "exit":
        data.append(user_input)

print("You entered:")
for item in data:
    print(item)
```

🔎 Explanation:

- Loop runs until user types `"exit"`
- We store values in list
- After loop ends → print all stored values

🔷 Question 2: Print All Odd Numbers from 1 to 100

📌 Question:

Write a program to print all odd numbers between 1 and 100.

✅ Solution (Using for loop)

```
for i in range(1, 101):
    if i % 2 != 0:
        print(i)
```

🔎 Logic:

- Odd numbers give remainder 1 when divided by 2
- `range(1,101)` → 1 to 100

🔷 Question 3: Find Factorial of a Number

📌 Question:

Write a program to calculate factorial of a number using while loop.

✅ Solution

```
num = int(input("Enter a number: "))
fact = 1
i = 1

while i <= num:
    fact = fact * i
    i += 1

print("Factorial is:", fact)
```

🔎 Example:

If user enters 5
Factorial = 5 × 4 × 3 × 2 × 1 = 120

🔷 Question 4: Sum of First N Natural Numbers

📌 Question:

Write a program to calculate sum of first N natural numbers.

✅ Solution (Using for loop)

```
n = int(input("Enter value of n: "))
total = 0

for i in range(1, n+1):
    total += i

print("Sum =", total)
```

🔎 Formula Concept:

Sum of first 5 numbers → 1+2+3+4+5

🔷 Question 5: Reverse a Number Using While Loop

📌 Question:

Write a program to reverse a number.

✅ Solution

```
num = int(input("Enter a number: "))
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

print("Reversed number:", reverse)
```

🔎 Example:

Input: 123
Output: 321

✅ PART 2: 2 OUTPUT PREDICTION QUESTIONS

🔷 Question 6: Guess the Output

```
for i in range(1, 4):
    for j in range(1, 3):
        print(i, j)
```

✅ Output:

```
1 1
1 2
2 1
2 2
```

```
3 1
3 2
```

🔎 Why?

Outer loop runs 3 times
Inner loop runs 2 times each time
Total = 3 × 2 = 6 outputs

🔷 Question 7: Guess the Output

```
i = 1
while i < 5:
    if i == 3:
        break
    print(i)
    i += 1
```

✅ Output:

```
1
2
```

🔎 Why?

When i becomes 3 → break stops loop immediately.

✅ PART 3: 1 DEBUGGING QUESTION

🔷 Question 8: Find the Error

```
i = 1
while i <= 5
    print(i)
    i = i + 1
```

❌ Error:

Missing colon ： after while condition.

✅ Correct Program:

```
i = 1
while i <= 5:
    print(i)
    i = i + 1
```